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Unit 1 - Creating Different Colors

PDF Documents

PDF Document 1: 1738217867597-Chapter 5 - Unit 1 - Creating Different Colors.pdf

This PDF document is included in the unit materials.

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UNIT 1 CREATING DIFFERENT COLORS

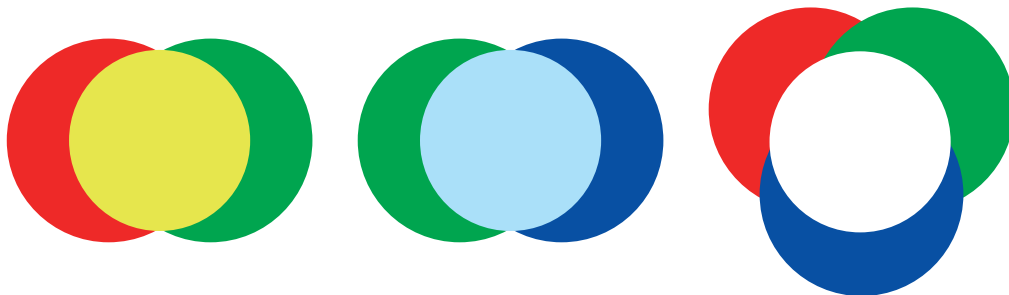
●● How Many Colors Are There in the World? _____



Although it feels like the world is entire of countless colors, nearly every color can be created using just three primary colors: Red (R), Green (G), and Blue (B). Combining these three colors in different ways can produce a wide range of colors. This principle is used in screens and lighting to display various colors.

EXAMPLE

Mixing red and green creates yellow, and mixing green and blue creates cyan. When red, green, and blue are combined at full brightness, we get white. Adjusting the combination and brightness of these three colors allows us to create a vast spectrum of colors.



This principle works because human eyes have cells sensitive to red, green, and blue light. Thanks to these cells, the RGB color model can represent nearly all our perceived colors.

● How Many Colors Are There in the World? _____

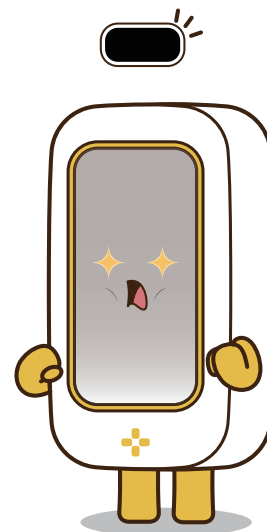
Digital devices like computers, TVs, and smartphones use the RGB concept to display colors. If you look closely at a screen, you'll see countless tiny dots or pixels. Each pixel adjusts its red (R), green (G), and blue (B) lights to produce specific colors.

Each color's brightness can be set from 0 to 255, allowing pixels to create various colors. Together, these pixels form the images we see.

EXAMPLE

Setting both red and green to 255 produces yellow. Setting red, green, and blue all to 255 results in white. When all three colors are set to 0, it makes black.

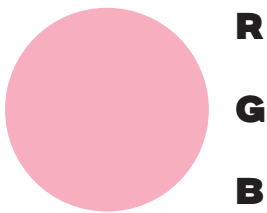
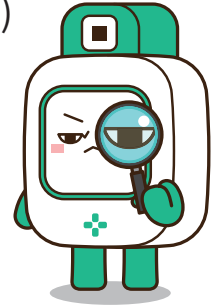
Computers can display millions of colors through RGB combinations. All the colors on a digital screen are created by blending red, green, and blue.



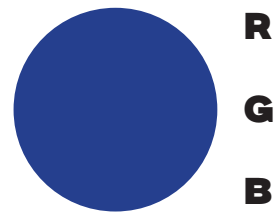


Activity

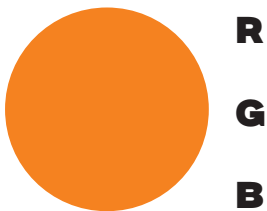
Determine the color combinations and their proportions (0 to 100) needed to create the colors below.



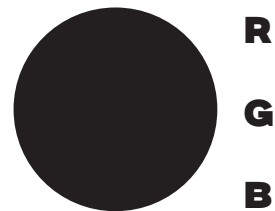
R
G
B



R
G
B



R
G
B



R
G
B